

POCKETSMART AI

Project Testing Phase

Purpose: The testing phase checks whether PocketSmart AI works correctly, gives expected results, and provides a smooth user experience.

1. Functional Testing

- Check whether all main features work correctly.
- Test adding, editing, and viewing income and expenses.
- Check budget and savings calculations.
- Verify that AI-based suggestions are generated correctly.

2. Input Testing

- Test valid user inputs.
- Test empty fields and incorrect values.
- Check how the system handles invalid data.
- Make sure the application does not crash because of wrong input.

3. User Interface Testing

- Check whether all buttons, fields, menus, and pages work properly.
- Make sure the information is displayed clearly.
- Check that the interface is simple and easy to understand.
- Test the application on different screen sizes where possible.

4. AI Response Testing

- Check whether the Gemini AI connection works.
- Verify that the AI receives the required information.
- Check whether responses are relevant to the user's input.
- Handle API errors or unavailable responses safely.

5. Calculation Testing

- Verify total income and total expense calculations.
- Check remaining balance calculations.
- Test budget limit calculations.
- Verify savings-related calculations with different values.

6. Error Handling

- Check API key or configuration errors.

- Test missing or incomplete input.
- Display understandable error messages.
- Make sure errors do not expose sensitive information.

7. Security Testing

- Check that API keys are not displayed in the application.
- Keep sensitive configuration in environment variables.
- Avoid displaying unnecessary personal financial information.
- Check that user data is handled safely.

8. Test Cases

Test Case	Expected Result
Add expense	Expense is saved and displayed
Add income	Income is recorded correctly
Calculate balance	Correct remaining amount is shown
Set budget	Budget value is accepted and displayed
AI assistance	Relevant AI response is displayed
Invalid input	Clear error or validation message is shown

9. Final Testing

- Run the complete PocketSmart AI application.
- Test all modules together.
- Fix identified errors and retest the application.
- Confirm that the final version is ready for demonstration.

Conclusion

Testing ensures that PocketSmart AI is reliable, user-friendly, and working as expected. After successful testing and correction of errors, the application can be prepared for final demonstration and deployment.